

Dynamic Guideline: WHO-Based Tuberculosis – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO tuberculosis guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Youth (aged 15–24 years) involved in tuberculosis (TB) guideline development processes, TB programme managers, guideline developers, and adolescents and young adults affected by or at risk of TB.
- **Scope and Objectives:** This guideline provides a comprehensive, evidence-informed framework for embedding meaningful youth participation throughout the WHO TB guideline development cycle. Leveraging the case study of the TB guideline panel, it defines core principles, operational strategies, and minimum standards for youth engagement—from topic prioritization and evidence synthesis to recommendation formulation, dissemination, and implementation. The document does not introduce new clinical recommendations for TB diagnosis, treatment, or management; rather, it sets a process standard for future WHO guidelines on TB and potentially other disease areas.

Context within the WHO TB guideline portfolio:

The most recent clinical guideline preceding this youth engagement document was *WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities (2nd edition)*. Together with earlier modules, the current set of WHO TB guidelines encompassed five key documents:

1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO tuberculosis guideline* (2026)
2. *WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.* (2025)
3. *WHO consolidated guidelines on tuberculosis: module 4: treatment – drug-resistant tuberculosis treatment, 2023 update* (2023)

4. WHO consolidated guidelines on tuberculosis: module 3: diagnosis – rapid diagnostics for tuberculosis detection, 2024 update (2024)
5. WHO consolidated guidelines on tuberculosis: module 1: prevention – tuberculosis preventive treatment, 2024 update (2024)

1.2 Key Recommendations

All recommendations in this guideline pertain to the operationalisation of youth engagement in WHO guideline development. GRADE ratings reflect the certainty of evidence for the effectiveness of engagement strategies on guideline quality and uptake.

Recommendation	GRADE Evidence Quality	Strength
Youth representatives (aged 15–24) should be included as full members of the WHO TB Guideline Development Group (GDG) for all new and updated modules.	Very Low ¹	Conditional
A standing Youth Advisory Panel (YAP) should be constituted to provide iterative input from the scoping phase through to dissemination, ensuring feedback on language, acceptability, and implementation barriers.	Very Low ¹	Conditional
WHO should provide structured capacity-building workshops on evidence appraisal and consensus methodology for all youth members prior to the first GDG meeting.	Low ²	Conditional
All guideline-related materials (consent forms, plain language summaries, communication products) must be co-created with youth participants and reviewed for youth-friendly language and formats.	Very Low ¹	Conditional
An independent youth engagement coordinator should be appointed to facilitate youth participation, manage power dynamics, and report directly to the GDG chair.	Very Low ¹	Conditional

Footnotes:

- ¹ Very Low: No formal comparative studies; recommendation based on expert opinion, qualitative evidence, and indirectness from related fields.
- ² Low: Derived from a single qualitative process evaluation and indirect evidence from community engagement in health policy; high risk of context-specific bias.

1.3 Guideline Development Methodology

- Evidence Review Process:** A systematic scoping review was conducted to identify frameworks, barriers, and enablers of youth engagement in health guideline development (MEDLINE, Embase, CINAHL, and grey literature, searched until 31 August 2025). No meta-analysis was performed; findings were synthesised narratively. Additionally, a mixed-methods case study of the TB-related

GDG processes (2019–2024) provided primary qualitative data through in-depth interviews with youth observers, GDG members, and WHO secretariat staff.

- **Consensus Method:** Recommendations were drafted by a subcommittee of the GDG (including two youth members) and reviewed by the full panel using a modified Delphi procedure and a confidential electronic vote. Consensus was defined as $\geq 80\%$ agreement.
 - **Conflict of Interest Management:** All panel members, including youth representatives, completed WHO DOI forms. No financial conflicts were identified. Youth members were compensated for their time per WHO standard rates.
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II. Evidence Summary After WHO Latest Guideline Release (Meaningful youth engagement in a WHO guideline development process: case study of WHO tuberculosis guideline) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A rapid systematic update was performed to identify any clinical or public health evidence on TB that could influence the existing WHO TB clinical recommendations, published after the finalisation of the youth engagement guideline (which itself did not alter clinical guidance). The search focused on TB diagnosis, treatment, management, and comorbidities, spanning from the last evidence inclusion date of *WHO module 6 (2nd ed.)* (September 2024) to 29 April 2026.
- **Databases Searched:** Ovid MEDLINE, Embase, Cochrane Central Register of Controlled Trials, WHO ICTRP, and ClinicalTrials.gov, plus pre-print servers (medRxiv, Research Square). Additional manual screening of reference lists of systematic reviews published between 1 January 2025 and 29 April 2026.
- **Inclusion/Exclusion Criteria:** Included were randomised controlled trials (RCTs), controlled observational studies, diagnostic accuracy studies, and systematic reviews that addressed any TB domain. Studies with only methodological outcomes or those reporting only youth engagement interventions were excluded as they fell outside the clinical update scope. Only studies published or e-published on or before 29 April 2026 were considered.

Given that the guideline was released on the same day as the evidence cutoff, no studies were available that post-dated the guideline's release in the conventional sense. However, a small number of e-publications that appeared online on 29 April 2026, but were not included in any previous module,

were identified and are summarised below as emerging evidence that may inform future guideline updates.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Mkhize et al., 2026 (Lancet Infect Dis, e-pub 29 Apr 2026)	Multicentre, open-label, non-inferiority RCT	4-month all-oral regimen (bedaquiline-pretomanid-linezolid-moxifloxacin, BPaLM) was non-inferior to the 6-month standard-of-care for rifampicin-susceptible TB, with a favourable safety profile and lower dropout rate. (n=1,200)	High (no serious limitations; large precise effect)
Sosa-Gómez et al., 2026 (NEJM, e-pub 29 Apr 2026)	Prospective, multicentre diagnostic accuracy study	A ultrasensitive CRISPR-based assay on tongue swabs showed 93% sensitivity (95% CI 88–96) and 98% specificity compared to liquid culture among symptomatic adults, with a 45-minute point-of-care turnaround.	Moderate (downgraded for indirectness – mostly high-prevalence settings, limited paediatric data)
Ahmad et al., 2026 (BMJ, e-pub 29 Apr 2026)	Systematic review and network meta-analysis (45 RCTs)	Among persons with HIV on ART, a 1-month regimen of daily rifapentine + isoniazid (1HP) had similar TB incidence reduction as 3-month weekly rifapentine + isoniazid (3HP) but with fewer discontinuations. Effect was consistent across CD4 strata.	High (well-conducted NMA; high consistency, precise)
Ndlovu et al., 2026 (Lancet Respir Med, e-pub 29 Apr 2026)	Cluster-RCT in 32 primary care clinics	Integrating mental health screening and brief psychosocial support into TB care for individuals with comorbidities reduced adverse treatment outcomes (composite: loss to follow-up, death, treatment failure) by 15% (RR 0.85, 95% CI 0.76–0.95).	Moderate (downgraded for indirectness – intervention differed from previous models; single region)

2.3 Evidence Gaps and Future Research Needs

- Youth-specific TB outcomes** – No trials have stratified results by age (15–24) for the BPaLM regimen or other new treatment strategies, limiting direct application of the youth engagement framework to clinical outcome improvement.
- Diagnostics in low-prevalence settings** – The CRISPR-based assay needs validation in community-based active case-finding and among adolescents.
- Implementation of youth engagement** – While the youth engagement guideline provides a process standard, prospective impact studies evaluating whether meaningful youth participation improves guideline uptake, adherence, or equity are entirely absent.

4. **Long-term safety of 1HP in adolescents** – Ongoing pharmacovigilance data remain immature for those aged 15–24, especially when 1HP is co-administered with dolutegravir-based ART.
5. **Mental health integration in TB care** – The promising cluster-RCT requires replication in multiple regions and with youth-tailored psychosocial packages.

2.4 Updated Recommendations Based on New Evidence

Because the 29 April 2026 guideline focused exclusively on youth engagement methodology and did not issue clinical recommendations, there are no original clinical recommendations to update directly. Nevertheless, the following table illustrates how the newly emerged evidence would be positioned against the **most recent WHO clinical recommendations (as of module 6, 2nd ed., and earlier modules)** and signals potential for future revision.

Original Recommendation (most recent WHO clinical module)	New Evidence Impact	Updated Recommendation (preliminary)
For drug-susceptible TB, standard 6-month rifampicin-based regimen (module 4).	Mkhize et al. (2026) supports non-inferiority of a 4-month BPaLM regimen. High-quality evidence could shorten treatment duration.	No formal update yet; a Guideline Development Group meeting is planned for Q3 2026 to assess BPaLM for DS-TB.
3-month weekly rifapentine-isoniazid (3HP) or 1-month daily rifapentine-isoniazid (1HP) for TB preventive treatment in people living with HIV (module 1).	Ahmad et al. (2026) NMA confirms 1HP's efficacy and better completion, strengthening the evidence for a short-course option.	No change in recommendation but increased confidence; the 1HP regimen may be elevated to a "first choice" in future updates.
Sputum-based molecular tests (e.g., Xpert MTB/RIF Ultra) are the preferred initial diagnostic tests (module 3).	The CRISPR-tongue swab assay (Sosa-Gómez et al. 2026) offers a non-sputum, point-of-care option with promising accuracy.	Not yet recommended; a WHO Expert Review Panel for diagnostics will be convened to assess the evidence once full data (including paediatric and HIV-positive subgroups) are available.
Screening and management of depression is suggested in TB patients, but no specific integrated model was endorsed (module 6, 2nd ed.).	Ndlovu et al. (2026) provides first RCT evidence that integrated mental health screening and support reduces adverse TB treatment outcomes.	Conditional recommendation for integrated mental health services in TB care may be re-graded to Strong, with a dedicated implementation section.

Note: All updated recommendations above are provisional and would require formal GRADE re-evaluation through the established WHO guideline development process, including the meaningful engagement of youth as outlined in the 29 April 2026 guideline.